

Summary of Backpropagation-Free Training

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1 Introduction

Backpropagation-free (BP-free) training represents an innovative approach in deep learning, where the traditional gradient calculation steps of backpropagation are eliminated. This method offers the potential to reduce training time, as it no longer requires the time-consuming process of backpropagation. Additionally, BP-free training can reduce memory usage since there is no need to store the intermediate results from forward propagation for gradient calculation. In the following, I will summarize the key insights from [1], [2], and [3], discussed in Sections 2, 3, and 4, respectively, before sharing my perspectives on future work in Section 5.

2 A Survey of Backpropagation-Free Training for LLMs

This paper categorizes BP-free training into three types: perturbed models, perturbed inputs, and no perturbation. The following provides a detailed discussion of each approach.

2.1 Perturbated Model

The perturbated model approach can be further divided into three categories: forward gradient, zeroth-order optimization, and evolution strategy. Their common characteristic is adding perturbations to the model parameters to estimate gradients.

2.1.1 Forward Gradient

The forward gradient method computes the gradient by calculating the Jacobian-vector product during forward propagation. This approach requires significant computational effort during the forward pass to obtain the gradient, which leads to relatively high accuracy in gradient estimation. However, the trade-off is that this method does not reduce the training time.

2.1.2 Zeroth-order Optimization

Zero-order optimization, also known as gradient-free optimization, differs from gradient-based methods in that it does not rely on the gradient of the objective function. Instead, it seeks to find the optimal solution by sampling or directly evaluating the objective function. While zero-order optimization does not compute gradients explicitly, it often uses numerical differentiation to approximate them. Numerical differentiation employs finite difference methods to estimate the gradient, which then helps guide the optimization process.

2.1.3 Evolution Strategy

Evolutionary Strategy (ES) is a population-based heuristic search algorithm, similar to genetic algorithms and evolutionary computing, primarily used in scenarios where the model gradient is inaccessible.

2.2 Perturbed Input

The perturbed input method involves introducing perturbations to the input and performing two forward propagations: one with the original input and another with the perturbed input. The model parameters are then updated based on the responses of the hidden layers to both types of forward propagations. The Forward-Forward Algorithm (FF), a renowned DP-free algorithm, is developed using this perturbed input approach.

2.3 No Perturbation

Some backpropagation-free methods [4–6], do not incorporate perturbation. However, these algorithms remain relatively independent and have not seen widespread application.

3 One Forward Is Enough for Neural Network Training via Likelihood Ratio Method [2]

This paper introduces the Unified Likelihood Ratio (ULR) method, a novel approach to neural network training that requires only a single forward pass. The ULR method aims to overcome the limitations of traditional backpropagation (BP) by improving the efficiency of the training process while also increasing the design freedom of neural network architectures.

3.1 ULR Method for Gradient Estimation

The ULR method estimates gradients by injecting noise into the weights or logits of neurons. During training, ULR generates noise that follows a specific distribution for each layer or logit and computes the network’s loss. It then compares the losses, incorporating noise information, and employs expectation algorithms and variance reduction techniques to estimate the gradients. These gradients are subsequently used to update the network weights.

3.2 Key Benefits of the ULR Method

The ULR method is highly adaptable and can be applied to different models and gradient updating methods. First, it is compatible with a variety of neural network architectures. This paper demonstrates the effectiveness of the ULR method across different models, including convolutional neural networks (CNNs), recurrent neural networks (RNNs), and graph neural networks (GNNs). Second, the ULR method is highly scalable, as it can be integrated into the backpropagation training process as a gradient calculation technique, rather than requiring the design of a separate gradient update method.

4 Fwdllm: Efficient Fedllm Using Forward Gradient [3]

This paper introduces FwdLLM, an innovative federated learning protocol designed to improve the training efficiency of large-scale language models (LLMs) on mobile devices. The core innovation of FwdLLM lies in adopting a backpropagation-free training method, where perturbation inference is performed instead of traditional backpropagation, significantly reducing memory usage and computation time.

While federated learning helps maintain data privacy on mobile devices, traditional backpropagation methods face performance bottlenecks when training large models. FwdLLM can address the performance limitations of mobile devices, but it also faces several challenges. It is designed to tackle the following three main challenges:

- **Effectiveness of the Backpropagation-Free Method:** Traditional backpropagation-free methods perform well on small models, but on large models, more perturbation inferences are required. FwdLLM combines backpropagation-free training with efficient parameter training techniques, enabling it to be effective even in LLMs.

- **Dynamic Allocation of Computational Load:** When training is performed across multiple devices, how to appropriately distribute the computational load is a challenge. FwdLLM systematically and adaptively allocates computational tasks to achieve a balance between convergence speed and accuracy, thereby optimizing the overall training performance.
- **Effective Selection of Perturbation Samples:** Heterogeneous perturbations can impact model convergence. FwdLLM uses a discriminative sampling mechanism to select perturbation prediction samples that are more valuable for model convergence, improving both the efficiency and effectiveness of model training.

With the above design, FwdLLM demonstrates significant advantages in experiments, including up to three times faster convergence speed and up to 14.6 times reduced memory usage. This makes it possible to fine-tune billion-parameter models (such as LLaMA) on standard commercial smartphones, breaking previous technological limitations.

5 Prospect of Backpropagation-Free Training

Deep learning based on the Unified Likelihood Ratio (ULR) shows great potential, as it can be integrated with existing training methods that rely on backpropagation-computed gradients. Moreover, it has demonstrated adaptability across a wide range of models. A promising research direction could be combining ULR with distributed learning to enhance its robustness against uneven data distributions. This can be achieved by modifying the computational gradient steps in ULR to better handle such disparities.

References

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